Sci-Fi Spark: A Human-AI Co-Creation System for Science Fiction Ideation

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Abstract

In science fiction writing, ideation demands both novelty to construct fictional worlds and consistency to maintain internal and temporal logic within those worlds. While large language models (LLMs) are increasingly adopted as co-creators, generated ideas often lack surprise and struggle to maintain consistency. Moreover, current interaction paradigms of human-AI co-creation systems fail to support the fragmented and iterative nature of science fiction ideation. To address these challenges, we introduce Sci-Fi Spark, a human-AI co-creation system to support inspiration and organization in the ideation phase. The system features an Ideation Canvas for visualizing relationships between fragmented ideas, a Novelty Generator that applies computational creativity strategies to produce novel worldbuilding inspirations, and a Consistency Generator to produce context-aware storyline suggestions. A technical evaluation and a user study with writers show that Sci-Fi Spark enhances both novelty and consistency, while enabling iterative co-ideation with LLMs.

CCS Concepts

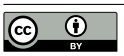
• **Human-centered computing** → **Interactive systems and tools**.

Keywords

Science fiction, Human-AI co-creation systems, Creativity support

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1 Introduction

Speculative fiction constructs stories in fictional worlds to reflect on societal, cultural, or technological possibilities, including genres such as science fiction and fantasy [63]. In the ideation phase of speculative fiction writing, writers have to achieve both novelty and consistency in their stories [67, 74]. Novelty is required to imagine fictional worlds that depart from reality, while consistency is required to ensure that story development adheres to the logic within those worlds [53, 90].

Recently, large language models (LLMs) have been adopted as co-creators in fiction writing, assisting with tasks such as plot development [82] and story infilling [12, 71, 86]. However, LLM-generated texts often lack novelty, frequently relying on "clichés" [21, 56], and struggle to maintain consistency [59, 81]. While domain-specific strategies have been explored to enhance novelty such as metaphor-based generation for poetry [46], few works address how to improve novelty in worldbuilding-intensive domains such as speculative fiction. For consistency, most approaches focus on optimizing generation of outputs [18, 50, 87] but ignore the writer's role in the co-creative process. This limits writers' control and flexibility, which are critical in speculative fiction ideation where iterative revisions are frequent. As such, it remains unclear whether domain-specific generation strategies for novelty and output-focused approaches for consistency can adequately support the novelty and consistency demands of speculative ideation.

To address these gaps, we target ideation in speculative fiction, with a particular focus on science fiction (Sci-Fi) as a representative case because the pursuit of cognitive estrangement in Sci-Fi makes the novelty and consistency demands especially salient [19]. To understand the underlying factors behind these gaps from the perspective of Sci-Fi writers and directly guide the system design, we conducted a formative study. Writers reported their ideation workflows as alternating between worldbuilding that involves imagining settings of the fictional worlds, and storylining that involves

constructing trajectories of story events within those worlds. Together, these two activities form an iterative workflow. Writers attributed the above gaps in supporting such ideation workflows to limitations at the interaction level rather than model level. They found that current interaction paradigms obscure the structure and evolution of their ideas. On the one hand, LLMs often overfit to previous context input, generating predictable inspirations that hinder novel worldbuilding. On the other hand, the lack of support for representing and managing evolving story event trajectories limits the context-aware generation in storylining.

To explore how humans and AI systems can collaborate more effectively in Sci-Fi ideation, we propose the following research questions:

RQ1: How can we provide novel inspirations for worldbuilding?

RQ2: How can we help writers represent and manage consistency across storyline trajectories?

RQ3: How can we design a system that supports the nonlinear and iterative nature of Sci-Fi ideation?

To address these questions, we developed Sci-Fi Spark, a human–AI co-creation system for Sci-Fi ideation that consists of three components: the Ideation Canvas, Novelty Generator, and Consistency Generator. The Ideation Canvas represents writers' ideation process using idea blocks as units, which can denote either worldbuilding elements or story events, and linking them through two types of consistency lines. This structured representation facilitates interaction by making consistency relationships visible and manipulable to both the writer and the LLM system. The Novelty Generator, based on the TVTropes database[1] and Boden's theory of computational creativity [16], provides novel inspirations for worldbuilding. The Consistency Generator uses an iterative generation framework to propose context-aware storyline trajectories based on the structured context represented on the canvas. Together, the three components form an open ideation system that supports nonlinear workflows by using idea blocks for fragmented expression, consistency lines for structured story relationships, and the two generators for adaptive creativity support. A technical evaluation using pairwise comparisons indicates that our generators significantly improve the novelty of worldbuilding suggestions and the consistency of storyline trajectories. In addition, a user study with twelve Sci-Fi writers further shows that the system fosters creativity through the generators, supports visualization and management of storyline trajectories via the canvas, and enables nonlinear, iterative workflows in human-AI co-creation.

In summary, this paper presents the following contributions:

(1) **An interaction paradigm for human-AI co-creation.** This work introduces the Ideation Canvas as a conceptual model that integrates temporal and internal consistency constraints into a unified visual structure. This structure establishes a shared context between users and LLMs, enabling the intuitive management of complex relationships. Consequently, the canvas offers a generalizable design metaphor for scaffolding human-AI co-creation tasks involving multidimensional narrative dependencies beyond the Sci-Fi domain.

(2) **A workflow-integrated operationalization of computational creativity.** This work integrates established computational creativity strategies into an interactive workflow to address the lack of novelty in LLM-assisted speculative fiction ideation. Leveraging the TVTropes database, we operationalize these strategies to combine trope attributes and transform narrative constraints to generate novel inspirations. This approach provides a methodological framework for mitigating the "cliché" tendencies of LLMs.

(3) **A human–AI co-creation system for Sci-Fi ideation.** We present Sci-Fi Spark as a system instantiation of the proposed paradigm that supports both worldbuilding and storylining. A technical evaluation validates the effectiveness of the proposed generation strategies, and the user study demonstrates improved support of nonlinear ideation workflows. These findings offer design implications for supporting complex, iterative processes in future human-AI co-creation systems.

2 Related work

2.1 Science fiction writing

Science fiction (Sci-Fi) emerged as a genre that facilitates critical reflection on societal and technological change, particularly throughout the twentieth century [64]. A defining feature of Sci-Fi is the central role of a novum, a speculative element that introduces cognitive estrangement [31, 37]. The novum differentiates Sci-Fi from other genres by enabling the construction of fictional realities beyond contemporary norms, which places a high demand on novelty [17, 53]. While novelty enables cognitive estrangement essential to Sci-Fi, consistency ensures that the fictional world remains plausible. The pursuit of novelty cannot rely on arbitrary imagination. The novum must be grounded in internal logic to remain credible [75]. In addition, the fictional world built around the novum must maintain strong consistency [51, 69, 89, 90], including coherent causal relationships across narrative arcs and chapter development [53]. In summary, Sci-Fi writing demands both high novelty and strong consistency to create cognitively estranging yet logically plausible fictional worlds.

2.2 LLMs for creative writing

LLMs trained on large-scale corpora have been widely used in creative writing due to their strong generation capabilities. To enhance novelty, prior studies have explored strategies such as suggestion generation based on user prompts [88], metaphor-based generation for poetry [38, 47], and audience-tailored suggestion generation [13]. LLMs have also improved novelty in plot development through character simulations [82] and story infilling based on user-provided context [12, 32, 44, 71, 80, 86]. Commercial tools such as Sudowrite [2], Writesonic [9], NovelAI [6], and ShortlyAI [1] integrate these capabilities by offering features such as style refinement and full narrative generation.

Empirical studies have sparked debates about LLMs' role in creativity. Some research shows that LLMs enhance individual creativity by aiding idea articulation [52]. However, other studies suggest they reduce the diversity of collective ideas, as shown in experiments on LLMs' effect on short story creativity [35] and group brainstorming [58]. This debate highlights that LLMs' impact on

[1] https://tvtropes.org/

creativity depends on the context. Some argue that LLMs are more effective when individual creativity is prioritized [52], while others note that task complexity matters: LLMs can inspire stimulation in simple tasks but may cause creative fixation in complex ones [27]. Recent research has further explored domain-specific creativity support through conceptual design assistance systems augmented by LLMs [22–24], highlighting the potential of LLM integration in facilitating complex creative tasks.

While these studies provide valuable insights into the broader role of LLMs in creative processes, most focus on general writing contexts, with limited attention to genres like Sci-Fi that place higher demands on novelty. Sci-Fi involves imagining unfamiliar, cognitively estranging worlds, making the generation of novel ideas challenging. In response, this work introduces novelty generation strategies tailored to support ideation in such high-demand genres.

In addition to novelty, LLM-generated text often lacks consistency and coherence [59, 81]. To address this, recent studies have explored two major types of solutions: (1) planning-based generation, which first creates high-level structures (e.g., outlines [81], plot skeletons [79, 85], or keyword sequences [43]) before expanding them into complete narratives, and (2) generate-and-rerank, which generates multiple candidates and selects outputs based on goal alignment or logical plausibility [11, 62]. These methods have improved consistency at the plot [87], stylistic [50], and character levels [18, 55, 91].

However, these solutions primarily optimize model outputs, overlooking the writer's active role in the creative process. Such linear generation workflows conflict with the nonlinear and iterative nature of ideation, limiting writers' control over generation. In response, this work introduces an interactive approach to support consistency generation through human-AI collaboration.

2.3 Visualization for writing

Before LLMs, narrative visualization tools primarily helped writers organize story structures. For example, Scrivener offers a corkboard-style interface for arranging scenes and chapters [8], while tools such as Plottr and Aeon Timeline visualize story progression through character and plot timelines [7, 10]. These tools externalize narrative flow, supporting writers in maintaining and reflecting on story development. However, they largely focus on single relationship types, typically temporal sequence or hierarchical structure, and lack the ability to represent complex and multiple narrative relationships.

With the introduction of LLMs, visual representations have increasingly functioned as communication media for human–AI co-creation [20]. Such communication spans both intent and context. At the intent level, recent systems use visual elements to express user intentions and interpret model outputs for controllable generation. For instance, TaleBrush conveys generation intent through user-sketched character fortune curves [29]; Patchview employs a dust-and-magnet visual representation to support sensemaking, steering, and correction of LLM-generated story-world elements [30]; and SARD enables users to construct graph-structured prompts of characters, actions, and relations that condition LLM-based generation [66]. These systems treat visualization primarily as an interface for articulating generation intent and interpreting model outputs. Beyond intent, context is equally critical in fiction writing and should also be accessible to LLMs for effective co-creation [54]. Several systems therefore focus on visualizing and manipulating narrative context, particularly for revision. For example, Visual Story-Writing transforms existing narrative text into coordinated entity–action, location, and timeline views, allowing writers to express revision intentions through direct manipulation [57]. Portrayal visualizes character-level information across a story to support reflection and refinement of character development [42]. In the commercial domain, Sudowrite utilizes a canvas interface for brainstorming, where writers can place scattered idea cards and outlines structured by predefined narrative templates [72]. The system leverages LLMs to generate these outlines, context-aware cards within the outline, or expand single cards into other possibilities.

Speculative fiction ideation supported by LLMs requires visualization to help writers express and manage multiple narrative relationships beyond simple timelines, including causal dependencies, and world rules [67]. Such multiple relationships must be maintained as a shared context for human–AI co-creation. Existing narrative visualizations typically handle a single relationship or localized intent and rarely integrate multiple narrative relationships as LLM-readable constraints for co-creative generation.

3 Formative study

To understand the workflows of Sci-Fi ideation and how Sci-Fi writers collaborate with LLMs during ideation, we conducted a formative study. While prior works broadly explored the roles of LLMs in co-creative writing, our study targets the ideation phase in Sci-Fi writing to directly guide system design.

3.1 Method

We recruited seven participants (P1 - P7) from an online Sci-Fi writing community to participate in a focus group and ideation task. Participants were aged 20 to 32 (M = 25.59, SD = 5.02) with one to nine years of Sci-Fi writing experience (M = 3.93, SD = 2.71). Three participants were expert writers with over five years of experience and formally published Sci-Fi novels through traditional publishers (excluding informal web-based publications), and four were novices without publications. All had prior experience using LLMs for writing. Detailed participant information is presented in Table 1.

Table 1: Participant information of the formative study.

ID	Gender	Age	Writing Experience	Written Sci-Fi Works (published)	AI Usage Frequency
P1	Female	28	1	3 (0)	Rarely
P2	Male	20	5	10 (7)	Weekly
P3	Female	32	6	17 (6)	Rarely
P4	Female	25	1	2 (0)	Weekly
P5	Male	20	3	2 (0)	Weekly
P6	Female	20	2.5	3 (0)	Weekly
P7	Female	32	9	9 (3)	Rarely

Note: Numbers in parentheses under "Written Sci-Fi Works" indicate formally published Sci-Fi works.

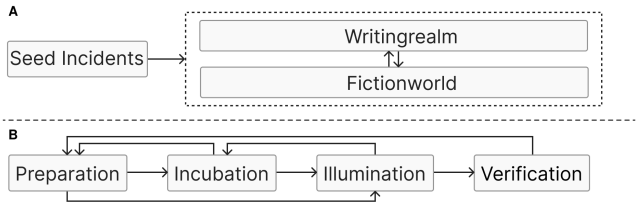


Figure 1: Two representative creative writing ideation models.

We summarized two general creative writing ideation models adapted from prior literature [36, 61]. The first model (Figure 1 A), adapted from Doyle [36], describes the ideation process as alternating between a "writingrealm", where writers apply storytelling knowledge, and a "fictionworld", where story elements emerge more intuitively. The second model (Figure 1 B), adapted from Wallas's seminal framework [77], outlines the classic four-stage creative process: preparation, incubation, illumination, and verification. Both models emphasize that creative writing ideation is iterative and nonlinear with writers often revisiting or skipping phases. These models served as discussion frameworks to help participants reflect on their own Sci-Fi ideation workflows during the focus group.

Participants first completed a brief background survey (5 minutes) on their writing experience and LLM usage. We then conducted a focus group session (60 minutes), where participants discussed how Sci-Fi ideation workflows align with or diverge from the provided models to identify Sci-Fi specific features. The whole session was audio-recorded, and participants received a compensation of 10 USD for their time.

After the focus group, participants engaged in an ideation task (60 minutes) using ChatGPT (GPT-4), the most commonly used LLM tool reported in the background survey. They were asked to develop and submit a short story outline following their discussed workflows while verbalizing their thoughts using a think-aloud protocol. To ensure thematic diversity and avoid overlap with public data, we provided three themes from ongoing Sci-Fi writing competitions:

(1) "Coexistence": Exploring the relationship between species, civilizations, and values in the future [3];
(2) "The Future State of the Stars": Imagining scenes from the near to distant future on Earth, within the solar system, or in other dimensions [4];
(3) "Building Yesterday's Future for Everyone": Revisiting the optimism and nostalgia of early 1960s Sci-Fi visions [5].

After the ideation task, participants completed a semi-structured interview (30 minutes) focused on their perceptions, challenges, strategies to address these challenges, and expectations regarding co-ideation with LLMs. All interviews were audio-recorded, and participants received 16 USD for their participation. For analysis, two researchers independently coded the transcribed recordings and iteratively refined themes through discussion until reaching consensus.

3.2 Findings

3.2.1 Sci-Fi ideation workflows. In our focus group, participants agreed that Sci-Fi ideation shares a similar structure with the general creative writing ideation models (Figure 1), but involves unique alternation between worldbuilding and storylining. Writers often begin with a seed idea inspired by daily observations or existing works, and refine it into a core concept. Worldbuilding was considered the core appeal of Sci-Fi. It includes imagined settings about scientific principles, social systems, and character personas. Participants emphasized that successful worldbuilding should produce a sense of cognitive estrangement [60], making novelty a key demand in this phase.

Storylining involves constructing story trajectories within the fictional world. Participants noted that Sci-Fi storylining often includes multiple layers, such as main plots, subplots, and character arcs. Participants stressed that consistency is key to ensuring credibility in imaginative settings, and identified two distinct types of consistency required in Sci-Fi ideation:

(1) **Temporal consistency:** Story events must follow a clear and continuous time order.
(2) **Internal consistency:** Story events must follow the established rules and logic of the fictional world.

Across both phases, participants typically used LLMs to articulate vague ideas and retrieve scientific knowledge. However, specific usage varied: in worldbuilding, LLMs were primarily leveraged to generate divergent inspirations for novel concepts; in storylining, they were used to bridge narrative gaps to maintain continuity.

3.2.2 Perceptions and challenges of co-ideation with LLMs. Participants generally recognized several benefits of using LLMs in the ideation process. First, LLMs were considered useful for retrieving scientific knowledge. Compared to conventional search engines, LLMs provided a more efficient and understandable way to obtain this knowledge. As P2, P3, and P6 noted, "I use it like an encyclopedia to help explain things." Second, LLMs helped enhance the articulation of ideas. For instance, P4 said, "ChatGPT generated ideas including the phrase 'conflict fusion model,' and that was the direction I wanted to go." Third, LLMs were seen as a tool for generating complementary ideas. As P1 explained, "Sometimes I worry about being stuck in my own thinking and ChatGPT can offer other perspectives." Despite these benefits, most participants viewed LLMs as an enhanced search engine due to their limited ability to meet novelty and consistency.

Challenge 1 (C1): Lack of novelty in worldbuilding. Participants found LLM-generated ideas were often literal and predictable, overly constrained by previous inputs. While prior research has shown that writers seek unexpected outputs to trigger creativity [86], participants reported that LLMs often remained tightly anchored to previous prompts, even when they intended to shift toward more divergent and surprising ideas. As P2 noted, "ChatGPT is like a knowledgeable reader. It picks up details I mentioned earlier, even when I don't want it to. But it's not a truly creative partner." In addition, the long-form outputs of LLMs were considered inefficient for ideation because participants preferred to extract keywords or key sentences for inspiration (P1–3, P5). Some participants (P4, P6, P7) further expressed concern that relying on predictable ideas might not only limit short-term outcomes but also weaken their creative abilities in the long run.

Challenge 2 (C2): Lack of consistency in storylines. Participants expressed that current LLM tools fail to maintain consistency in storyline ideation, especially in tracking evolving contexts. A major barrier appeared at communicating context to LLMs: both chat-based and document-based interfaces required writers to repeatedly describe their current context, which participants considered as time-consuming and cognitively demanding. As P2 noted, "I had to explain to ChatGPT every time how my setting had evolved." Participants also noted that such plain-text interfaces were insufficient for representing idea relationships. Some shared their experience of using canvas-based tools such as mind maps to represent fragmented ideas, but reported that these only supported single or ambiguous relationship types, and could not support the dual demands of internal and temporal consistency in Sci-Fi ideation (P5, P7).

3.2.3 Strategies and expectations for co-ideation. Beyond identifying challenges, we explored users' current coping strategies and their expectations for future system support.

Novelty: Seeking External Inspiration and Context Detachment. To combat "cliché" outputs, participants actively introduced constraints or external elements. Strategies included giving LLMs "many specific constraints and weird rules" (P3) or "teaching AI about niche topics" (P2) to force divergence. Consequently, to support such divergence natively, the majority ($N = 4$) expected that the system could temporarily detach from the narrative context. As P5 noted, they wanted to "choose to temporarily forget previous plots" to trigger "unexpected ideas." P6 further suggested "introducing external knowledge" to ensure diversity. These findings indicate that the system should facilitate external inspiration injection and allow for generation decoupled from the preceding narrative.

Consistency: From Manual Structuring to Deep Logic Support. To maintain consistency when conveying the established story context to the system, writers mainly relied on "summarizing context into a structured document" (P1) before feeding it to the AI. This cognitive burden led to a strong consensus on the need for visual structure tools. Participants explicitly requested the system to provide "mind map functions" to "visualize character personas" (P2) or "plot anchors" (P7) to track story nodes. These insights underscore the necessity of a visual interface to externalize narrative structure and an underlying mechanism to enforce consistency based on that structure.

3.3 Design goals

We translated the above challenges into four detailed design goals (DGs) to directly guide our system design:

DG1: Provide unexpected ideas to improve novelty in worldbuilding (C1). The system should inspire writers with unexpected and divergent ideas without being constrained by prior context, to support novel worldbuilding.

DG2: Provide context-aware ideas to improve consistency in storylining (C2). The system should generate ideas that align with the writer's context to maintain internal and temporal consistency in storylining.

DG3: Facilitate structured communication with LLMs (C2). The system should help writers efficiently convey evolving storyline trajectories to LLMs without disrupting their ideation workflows.

DG4: Support flexible and nonlinear workflows (workflow findings). The system should allow writers to freely create, modify, and reorganize ideas to support the iterative alternation between worldbuilding and storylining.

4 System design

Sci-Fi Spark is a human-AI co-creation system designed to support Sci-Fi writers in worldbuilding and storylining by providing novel and consistent inspirations through structured communication with LLMs. The system consists of: the Ideation Canvas (the left part of Figure 3 and 4), Novelty Generator (the right part of Figure 3), and Consistency Generator (the right part of Figure 4).

4.1 Ideation Canvas

To support clear visualization of writers' ideation processes and enable structured communication with LLMs (DG3), we designed the Ideation Canvas as the core interaction interface of the system. We adopted the canvas-based interface because it is well-suited for organizing fragmented ideas during the ideation phase [65]. As shown in Figure 2, the Ideation Canvas consists of three basic components: idea block, internal consistency line, and temporal consistency line.

(1) **Idea block.** An idea block is the basic unit on the canvas. We categorize idea blocks into two types: worldbuilding blocks and storyline blocks. Given the complexity of constructing sci-fi worlds, worldbuilding blocks are equipped with element tags, including World Rules, Character, Faction, Place, and Technology, which users can optionally apply for clear articulation. This taxonomy is adapted from Hergenrader's world catalogs [41] with genre-specific adjustments: we added World Rules to define the novum (speculative elements deviating from real-world laws central to sci-fi) [31] and replaced the Item category with Technology to emphasize the scientific nature of tools in sci-fi settings. Blocks may be created by the writer or suggested by the system's Novelty or Consistency Generator. This block format matches the fragmented nature of ideation and allows writers to freely capture both their own and system-generated ideas.
(2) **Internal consistency line.** Vertical lines indicate internal consistency between idea blocks. This top-down visual structure mirrors how people intuitively express constraints or dependency relationships [15]. For example, a vertical line from *idea block A* to *idea block B* indicates that *idea block B* depends on the settings or logic defined in *A*. Recognizing the multidimensional nature of internal consistency, the vertical lines are equipped with attribute tags representing five specific dimensions: Logical, Spatial, Societal, Psychological, and Thematic. Users can optionally apply these tags to clarify the specific dimension. This taxonomy is adapted from Ryan's narrative theory [68].
(3) **Temporal consistency line.** Horizontal lines indicate temporal consistency among idea blocks. This left-to-right visual structure resembles a timeline, making it intuitive for expressing temporal flow of stories [34]. For example, a horizontal line from *idea block C* to *idea block D* indicates that *D* occurs after *C*.

These basic components provide three key properties that support layered and structured communication during ideation: flexibility, transitivity, and composability as shown in Figure 2.

(1) **Flexibility.** Writers can use idea blocks independently to capture fragmented ideas without linking them. When needed, they can link the blocks either horizontally or vertically to express pairwise consistency.
(2) **Transitivity.** Linked idea blocks naturally inherit consistency from earlier ones. For example, in Figure 2, if I_0^0 is vertically linked to I_0^1, and I_0^1 to I_0^2, then I_0^2 follows the settings or logic of I_0^0. Similarly, horizontal links like I_0^0 –> I_1^0 –> I_2^0 construct a continuous sequence of stories.
(3) **Composability.** Internal and temporal consistency lines can be used together to express idea structures. Figure 2 illustrates all possible composability configurations when both the temporal consistency layer and internal consistency layer are set to 1. When used together, lower-level blocks inherit consistency relationships from higher-level blocks. As an example shown in Figure 2, I_0^0 links vertically to I_0^1 and I_1^1 (internal consistency), and I_0^1 links horizontally to I_1^1 (temporal consistency). Similarly, I_1^0 links to I_2^1 and I_3^1 in the same structure. I_0^0 also links horizontally to I_1^0. Together, this forms a timeline: I_0^1 –> I_1^1 –> I_2^1 –> I_3^1, where I_0^1 and I_1^1 follow the settings in I_0^0, and I_2^1 and I_3^1 follow the settings in I_1^0.

4.2 Novelty Generator

To provide unexpected worldbuilding inspirations that improve novelty (DG1), we proposed the Novelty Generator (Figure 3) based on Boden's theory of computational creativity [16]. This theory has been widely applied in computational creativity and shown to enhance creativity [26, 78]. It identifies three types of creativity strategies: combinational, explorational, and transformational. Among them, explorational and transformational creativity both operate within a conceptual space, but transformational creativity enables more radical innovation by breaking or extending the constraints of that system. To maximize the novelty of ideas, our system implements two generation modes based on combinational and transformational creativity strategies.

The two modes share the same interaction flow. Writers begin by selecting a worldbuilding idea to match a seed trope, optionally add requirements, and receive generated novel inspirations.

(1) **Target element selection.** For worldbuilding blocks, writers can optionally select target element tags (Figure 3E) to precisely control the generation. This ensures the generated inspirations explicitly includes the selected element categories.
(2) **Seed trope (Figure 3 F).** Writers can select any idea block on the Ideation Canvas (Figure 3 B) to initiate generation. Both combinational and transformational creativity strategies used for the generation require a well-defined concept to operate on [16], but writers' ideas during ideation are often brief or ambiguous [70]. So the system matches the selected idea block to the most semantically similar with the same element tags and concise concept within the scope of the specified element tags from a speculative fiction trope database (extracted from TVTropes [2]) as a seed trope. We use this database for its broad and structured coverage of speculative fiction tropes, but it could be replaced or extended in future iterations and is used here only as a demonstration resource. Our contribution lies in how the Novelty Generator transforms seed tropes into novel worldbuilding suggestions, not in the database itself. The matched seed trope then serves as the base in the combination mode or the conceptual space in the transformation mode.
(3) **Requirements (Figure 3 J).** Writers can optionally enter constraints or desired directions to guide the generation, such as "introduce a new species" or "set in an extreme environment".
(4) **Novel inspirations (Figure 3 K).** The generator produces novel worldbuilding inspirations as results. Writers can add satisfactory results to the canvas or regenerate after modifying the seed idea or requirements (Figure 3 L).

While the interaction flow is the same, the two modes differ in how worldbuilding inspirations are generated, drawing on recent work that employs generative models for interpreting and practicing computational creativity [25, 78].

(1) **Combination Mode.** Starting from the seed trope, the generator recommends an additive trope from the same database, automatically selecting within a moderate semantic similarity range to balance relevance and divergence [39]. Writers can refresh or manually browse the additive trope from the database list (Figure 3 G), allowing precise control over which worldbuilding ideas are combined. An LLM then operationalizes the combinational creativity following the two steps: (1) extracts key attributes from both tropes, and (2) synthesizes a new suggestion by logically blending them while avoiding shallow or redundant overlap. For example, combining "causal paradox" (time loop) with "symbiotic interface" (living systemship) might yield "time manipulation through biological perception feedback loops."
(2) **Transformation Mode.** In this mode, the seed trope is treated as a conceptual space. An LLM then operationalizes the transformational creativity following the three steps: (1) identify implicit constraints of the space (e.g., linear temporality, singular consciousness); (2) propose ways to break or extend those constraints (e.g., redefining physical laws or introducing collective cognition); (3) generate a new suggestion that remains coherent yet introduces a surprising conceptual shift. For instance, transforming "Layered World" may shift layers from spatial dimensions to cognitive ecosystems shaped by collective consciousness.

4.3 Consistency Generator

To generate consistent storyline trajectories that align with the writer's existing context (DG2), we developed the Consistency Generator (Figure 4). Unlike existing systems that require users to repeatedly describe context, our generator automatically infers context directly from the structured visualization on the Ideation Canvas. Writers begin by selecting a target idea block (Figure 4 G).

[2] https://tvtropes.org/pmwiki/pmwiki.php/Main/SpeculativeFiction

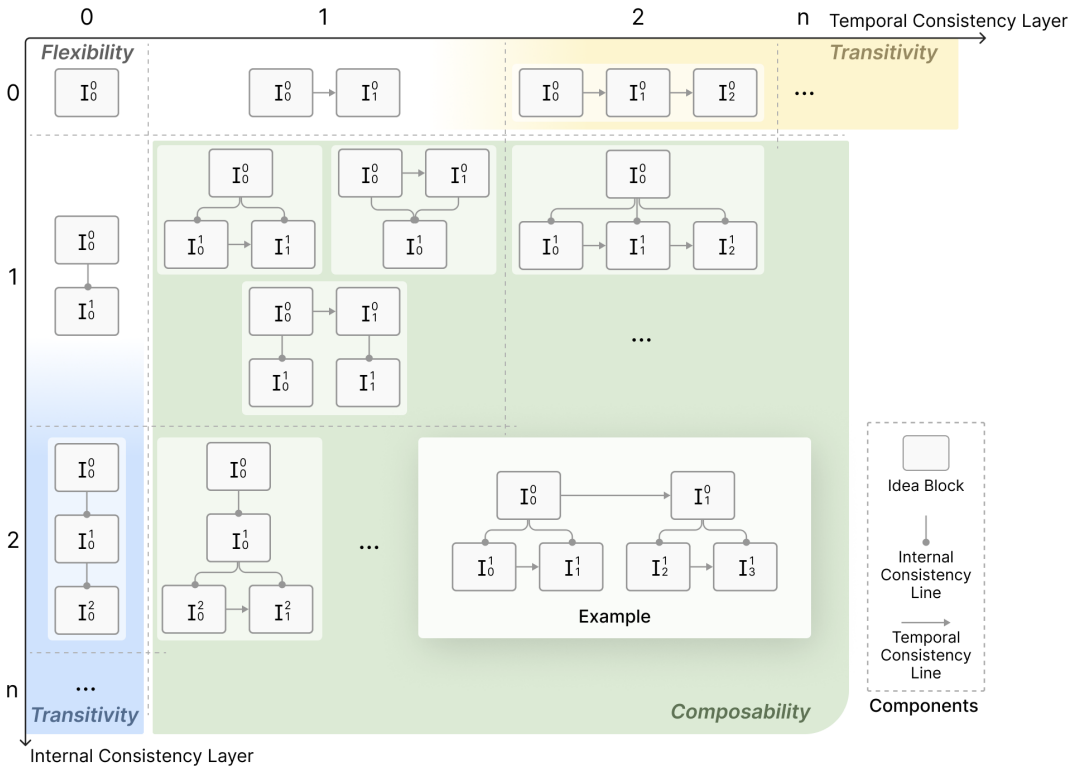


Figure 2: Visualization mechanism of the structures in the Ideation Canvas. It consists of idea block, internal consistency line, and temporal consistency line. The horizontal axis represents the temporal consistency layer, while the vertical axis represents the internal consistency layer. Idea blocks can be used either individually or linked in pairs (the "Flexibility" section). Idea blocks linked in the same way exhibit transitivity, where the consistency relationship is inherited across layers (the "Transitivity" section). The internal and temporal consistency relationships can be combined to form more complex structures (the "Composability" section).

The system then identifies surrounding context based on the block's vertical and horizontal consistency links. This approach reduces the cognitive load of manually explaining context to LLMs. For example, in Figure 4, if the selected *idea block G* is vertically linked to *idea blocks C*, *D*, and *E* and horizontally placed between *F* and *H*, the system will generate storyline suggestions that are internally consistent with *C*, *D*, and *E* and temporally positioned between *F* and *H*. To further refine the generation, writers can optionally apply attribute tags to specify the internal consistency dimension (Figure 4 I) or input requirements (Figure 4 K) to guide the generation toward their intention, such as "avoid violent outcomes".

To achieve these, the generator analyzes two types of links: temporal consistency and internal consistency to build a contextual frame for the target idea block. Based on this frame, ensuring that generated storyline suggestions align with the evolving storyline trajectory. To enhance reliability, the generator adopts an iterative refinement loop: generated storyline ideas are checked against inferred consistency requirements, and if they fail, the process regenerates with adjustments until satisfactory results are obtained.

The final verified storyline suggestions can be directly added to the Ideation Canvas as new idea blocks. By making consistency constraints explicit and leveraging them in generation, the Consistency Generator helps writers expand details, complete missing events, and create narrative bridges in storylining while maintaining coherence across both worldbuilding and plot development.

Together, these components and properties allow the Ideation Canvas to support structured representations of ideation. These structured representations can then be automatically interpreted by the systems to provide context-aware inspirations, as further detailed in Section 5. Thus, the Ideation Canvas supports efficient communication with LLMs without disrupting writers' ideation workflow.

4.4 Workflow

Sci-Fi Spark is a human-AI co-creation system designed to support nonlinear ideation workflows (DG4). Writers can engage in both worldbuilding and storylining, alternating freely between them as needed.

(1) **Worldbuilding.** Writers construct fictional settings, such as scientific principles, social systems, and character personas. They can record these ideas as worldbuilding idea blocks on

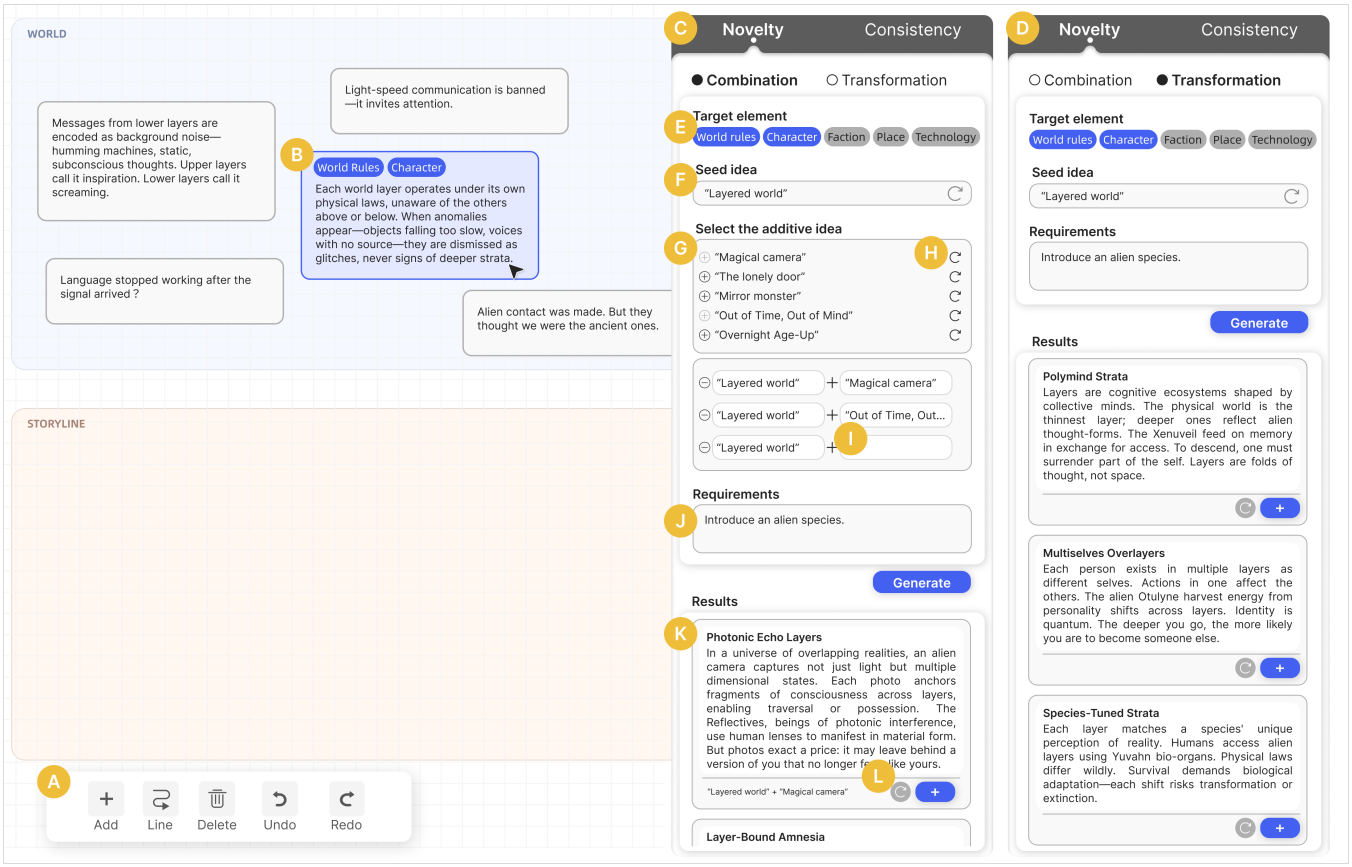


Figure 3: Sci-Fi Spark system interface with Novelty Generator in combination and transformation modes. The interface shown is translated from Chinese. It consists of the Ideation Canvas on the left and the Novelty Generator on the right. (A) The edit tools for the Ideation Canvas. (B) The selected idea block on the Ideation Canvas. (C) The Novelty Generator in combination mode. (D) The Novelty Generator in transformation mode. (E) Target element selectors to specify the required category for the generated worldbuilding idea blocks. (F) Seed idea that matches the selected idea block with an idea from a science fiction database. (G) Additive idea selection area, where the system recommends ideas from the same database. (H) Refresh button to update recommendations. (I) Manual browsing and selection of additive ideas. (J) Requirement input area for novelty generation. (K) Generation result area. (L) Controls to regenerate or add the results to the Ideation Canvas.

the Ideation Canvas and invoke the Novelty Generator to obtain divergent and novel inspirations.

(2) **Storylining.** Writers develop story trajectories within the fictional world. They can link idea blocks using consistency lines on the Ideation Canvas to represent story structures and invoke the Consistency Generator to get context-aware inspirations.

These two activities are supported by distinct generation strategies tailored to the writer's cognitive state. The Novelty Generator targets unconstrained blocks to foster divergent thinking, while the Consistency Generator targets contextually linked blocks to ensure convergent alignment. This design ensures that novel ideas remain unconstrained until the writer links them into the context, at which point the system shifts to enforcing consistency.

Ultimately, the system integrates three core mechanisms: idea blocks for fragmented expression, consistency lines for structured relationships, and generators for adaptive inspiration support. These mechanisms work together to form an open-ended ideation space rather than a linear generation pipeline, aligned with how Sci-Fi ideation unfolds in practice: through cycles of divergence, organization, and re-divergence [48].

5 Implementation

5.1 System architecture

Sci-Fi Spark adopts a decoupled architecture with a React front end [3] and a Django REST back end [4]. The front end renders the Ideation Canvas using React Flow [5] for interactive blocks and links, with Material UI components [6] for interface consistency. The back

[3] http://react.dev/
[4] https://www.djangoproject.com/
[5] https://reactflow.dev/
[6] https://mui.com/

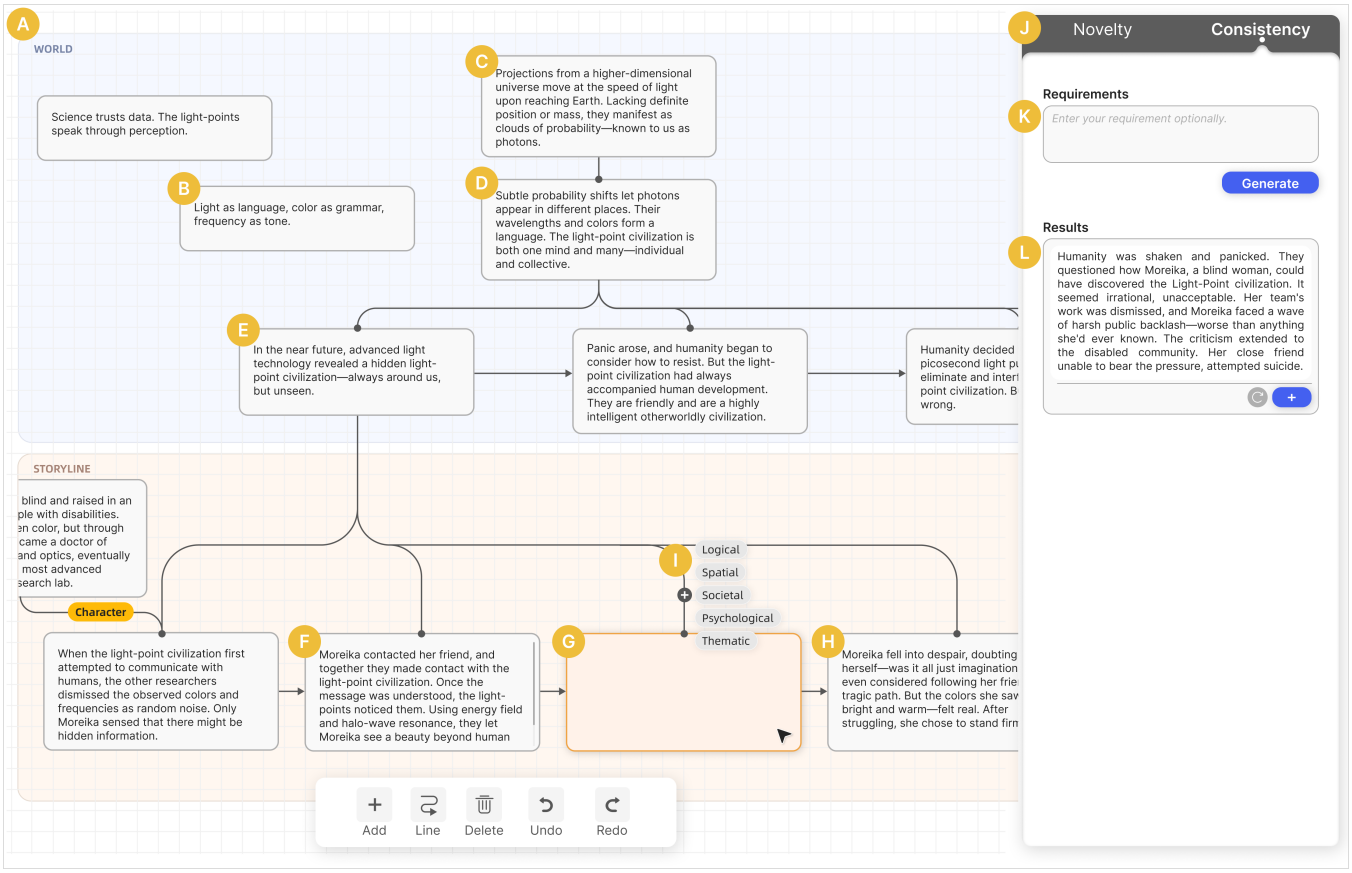


Figure 4: Sci-Fi Spark system interface with Consistency Generator. The interface shown is translated from Chinese. (A) The Ideation Canvas. (B) The fragmented idea blocks. (C) and (D) The settings of the fictional world. (E) One of the main story events. (F), (G), and (H) Three sub-story events that align with the main story event, with one in the middle selected for generation. (I) Attribute tags for internal consistency. (J) The Consistency Generator. (K) The requirement input area for consistency generation. (L) The result display area, where writers can regenerate ideas or add them to the Ideation Canvas.

end manages semantic retrieval, consistency reasoning, and text generation, and integrates with OpenAI [7] and Coze [8] APIs for LLM-based content generation.

5.2 Novelty Generator

The Novelty Generator (Figure 5 A) implements combination and transformation modes using GPT-4 with prompt engineering. User-selected idea blocks are semantically matched to a seed trope from a speculative fiction trope database (TVTropes [9]), retrieved via embedding-based similarity search combined with strict filtering by the user-selected element tags (if present). To enable this filtering, the database was pre-annotated with element tags (World Rules, Character, Faction, Place, and Technology) using GPT-4 via few-shot learning, followed by manual spot-checks to ensure accuracy.

Based on the retrieved seed trope, the system executes two distinct generation strategies. Combination mode selects an additive trope, also constrained by the selected element tag, within a similarity range of [0.4, 0.8], a range shown to balance relevance and diversity for stimulating creativity [39]. It then blends attributes of the additive and the seed trope. Transformation mode identifies implicit constraints and generates outputs by breaking or extending them. Crucially, when generating for tagged worldbuilding blocks, the prompt explicitly instructs the LLM to generate inspirations that incorporate corresponding element categories. To ensure genre fidelity, we incorporated core Sci-Fi criteria: *cognitive estrangement*, *scientific plausibility*, and *social implication* into the prompts for both modes, guiding the LLM to generate inspirations that are novel yet grounded in the Sci-Fi features [73]. We empirically set the generation temperature to 0.75 for Combination and 0.85 for Transformation [33, 49], balancing coherence with creativity. The generator returns a set of worldbuilding inspirations in JSON format, each consisting of a short title (≤ 15 Chinese characters) and a concise description (≤250 Chinese characters).

[7] https://platform.openai.com/
[8] https://www.coze.com/
[9] https://TVTropes.org/pmwiki/pmwiki.php/Main/SpeculativeFiction

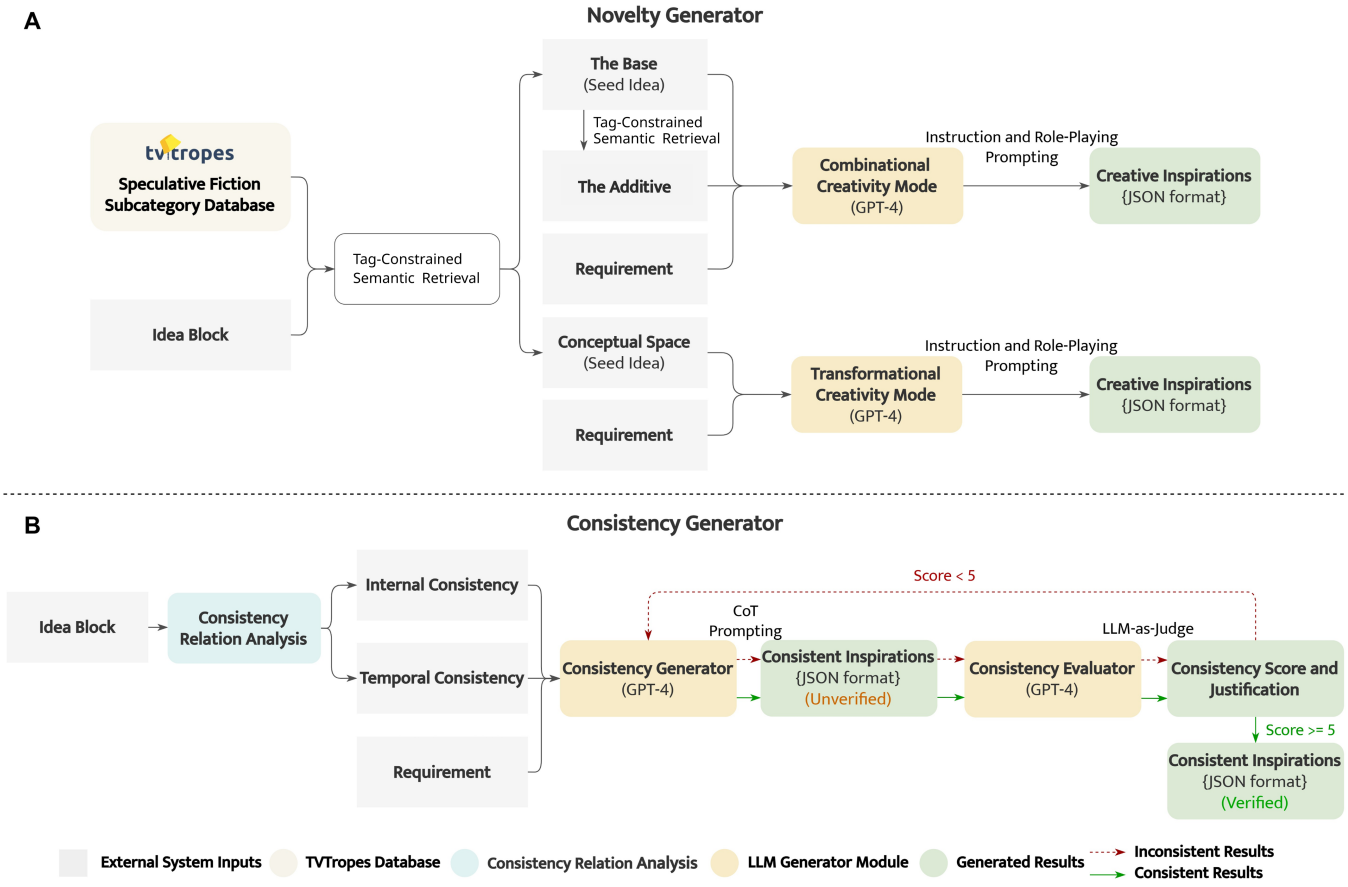


Figure 5: Technical implementation pipeline of the Novelty Generator (A) and Consistency Generator (B).

5.3 Consistency Generator

The Consistency Generator (Figure 5 B) also uses GPT-4, guided by a role-based prompt positioned as a Sci-Fi writing expert. This expert persona is grounded in the same Sci-Fi criteria (cognitive estrangement, scientific plausibility, and social implication) to ensure that the generated blocks respect the features of Sci-Fi. It analyzes temporal (horizontal) and internal (vertical) consistency links to construct contextual frames for a target block. If the internal consistency links carry attribute tags such as Logical or Spatial, these constraints are explicitly encoded into the prompt to direct the generation toward the specified consistency dimension. Generation follows a Chain-of-Thought (CoT) reasoning paradigm [83]. To enhance reliability, each output is evaluated by an LLM-as-Judge module [84], which scores temporal and internal consistency on a 1–7 scale. Outputs below the threshold (score < 5) trigger iterative regeneration until consistent results are obtained. This establishes an iterative framework that incrementally enhances consistency through refinement.

6 Technical evaluation

6.1 Method

To validate whether the Novelty Generator provides more novel inspirations and whether the Consistency Generator yields inspirations with better internal and temporal consistency, we conducted two pairwise comparison studies. In the novelty comparison, participants were asked to select the more novel inspiration ("Which text is more novel?"). In the consistency comparison, to assess performance both comprehensively and separately, participants were asked to choose according to three distinct dimensions: overall contextual fit ("Which text fits better in the context?"), temporal consistency ("Which text better bridges the timeline between events?"), and internal consistency ("Which text better aligns with the established world settings?").

In both studies, all evaluation materials were generated using GPT-4, with the same generator and input conditions. The only variable was the generation strategy, whether our proposed method was applied, to ensure that any observed differences were attributed to the strategy itself.

For the novelty comparison, we randomly selected 80 tropes from the speculative fiction category of the TVTropes database, and then

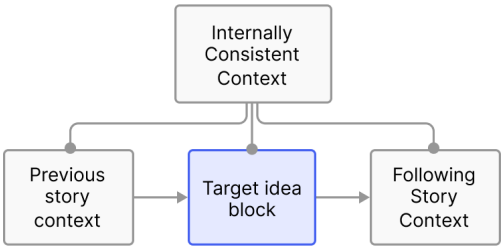


Figure 6: The context structure used in consistency comparison.

randomly assigned them into two groups: 40 for evaluating the combinational creativity strategy, and 40 for the transformational creativity strategy. For each trope, we generated one pair of novel inspirations using:

- Ours: Generate with our designed prompting strategy (combinational or transformational creativity strategy) from the trope.
- Baseline: Generate directly from the same trope without any creativity prompting strategy.

For the consistency comparison, we used the context structure shown in Figure 6, as it represents the minimal unit that jointly involves internal consistency and two forms of temporal consistency: "happens after" and "happens before" relationships. The content of each idea block was adapted from science fiction ideation results submitted by participants from the formative study, with their permission. A total of 38 samples were created by reformatting these submissions into the same structure with the target idea block masked for generation. Each sample was used to generate a pair of consistent inspirations of the target idea block: one using our method, and the other using the baseline. The only difference between the two methods was whether the iterative generation framework was used:

- Ours: Generate with our iterative framework.
- Baseline: Generate with a single-pass generator without evaluation.

We recruited 120 evaluators from a crowdsourcing platform Credamo[10], selecting those with an acceptance rate above 90%. We chose non-expert evaluators for the comparison tasks as the novelty and consistency of Sci-Fi works are ultimately judged by general readers. Of the 120 evaluators, 48 completed the novelty comparison: 24 evaluated combinational creativity and 24 evaluated transformational creativity, with each rating 10 pairs. The consistency comparison involved the remaining 72 evaluators, divided equally into three groups (n=24 each) to assess overall contextual fit, temporal consistency, and internal consistency separately. In these consistency comparison tasks, half of the evaluators rated 9 pairs, and half rated 10 pairs. We implemented mutually exclusive screening mechanisms to ensure that each participant could complete only one specific comparison task. Each comparison pair was rated by six independent evaluators to ensure reliability. One attention check question was embedded into each task set, and responses failing the check were discarded and replaced. All evaluators were compensated 2 USD.

We calculated statistical significance using one-sided binomial tests, and derived effect sizes (r) from subject-level Wilcoxon signed-rank tests. Inter-rater reliability was assessed via Fleiss' Kappa.

6.2 Results

Table 2: Pairwise comparison results for novelty and consistency. Novelty is evaluated under two creativity strategies, while consistency is assessed across three distinct dimensions.

Comparison Task	Strategy / Dimension	Ours	Baseline	Tie
Novelty	Combination	55.4%	32.5%	12.1%
	Transformation	56.3%	33.3%	10.4%
Consistency	Overall Contextual Fit	55.7%	36.0%	8.3%
	Temporal Consistency	61.4%	34.6%	4.1%
	Internal Consistency	66.3%	30.9%	2.8%

Table 2 summarizes the pairwise comparison results. Our method demonstrated significant improvements across all metrics ($p < 0.001$), with effect sizes ranging from medium to large.

6.2.1 Novelty Evaluation. Both combinational and transformational strategies produced significantly more novel inspirations than the baseline ($win\ rates > 55\%$, medium effect sizes $r \in [0.35, 0.42]$). Although inter-rater reliability was slight ($\kappa \approx 0.10$), typical of subjective creativity tasks, the robust statistical significance confirms a consistent preference for our method.

To understand the underlying factors, we analyzed representative examples of success and failure cases.

(1) A representative success case (combination strategy, 100% win): Under the "Henshin Hero" trope, the baseline offered a standard cliché of where the hero's change of form is merely "holographic bluff" . In contrast, our method introduced "Dimensional Tuning" which combines the process as a risky calibration with parallel realities. This novel combination created an outcome that evaluators found surprising yet plausible.
(2) A representative failure case (transform strategy, 100% loss): The attempt to maximize novelty resulted in cognitive overload. When generating an "Ancient Conspiracy," evaluators universally rejected our method's abstract inspiration of a "Quantum Consciousness Conspiracy" in favor of the baseline's more grounded "Dimensional Control Organization." The dense jargon in our method compromised narrative clarity, highlighting a delicate balance where our strategy effectively breaks fixations but must guard against excessive complexity.

6.2.2 Consistency Evaluation. Evaluators rated our method as significantly superior in internal consistency (66.3%, $r = 0.84$) and temporal consistency (61.4%, $r = 0.71$). Inter-rater reliability for these dimensions was $\kappa = 0.10$ and $\kappa = 0.16$ respectively. However, comparing these specific consistency metrics against the overall

[10]https://www.credamo.com/

contextual fit reveals a trade-off between logical rigor and narrative tension. While our method excelled in specific consistency dimensions, its advantage in overall contextual fit was comparatively smaller (55.7%, $r = 0.45$, $\kappa = 0.06$). A prime example is the story *Adaptation*, where the context required "bridging the discovery of an adaptive gene with subsequent human trials" . Our method generated a rigorous procedural bridge, "scientists designed a series of experiments... applying gene editing to volunteers" , which secured high temporal consistency scores (70%). In contrast, the baseline leaped immediately to a "bold plan for widespread application to solve resource shortages" . Although the baseline skipped the necessary experimental logic, it created stronger dramatic stakes. Evaluators appeared to prioritize this dramatic progression over the strict procedural correctness of our method, resulting in a lower overall preference (43%). This divergence underscores that overall contextual fit is influenced by narrative excitement as much as logical constraints.

7 User study

We conducted a user study with 12 participants to evaluate the effectiveness and user experience of Sci-Fi Spark. The study was approved by the university's institutional review board, and written informed consent was obtained from each participant. The study aims to answer the following questions:

Q1: How do writers perceive the novelty and contextual relevance of the generated inspirations?

Q2: How do writers perceive the usability of Ideation Canvas for visualizing and communicating idea relationships?

Q3: How does the system support writers' nonlinear and iterative ideation workflows?

7.1 Participants

We recruited twelve participants from an online Sci-Fi writing community in China who had not participated in the formative study. All participants were native Chinese speakers, consistent with the system language. Participants included four males and eight females, aged 20 to 39 (M = 24.17, SD = 5.20), labeled U1–U12. Participants had one to nine years of Sci-Fi writing experience (M = 3.71, SD = 2.62). Five participants were expert writers with formally published Sci-Fi works, while the other seven were novices without publication experience. All participants had prior experience using LLMs in writing. Detailed participant information is presented in Table 3.

Table 3: Participant information of the user study.

ID	Gender	Age	Writing Experience	Written Sci-Fi Works (published)	AI Usage Frequency
U1	Female	24	1.5 years	3 (0)	Monthly
U2	Female	23	5 years	8 (2)	Monthly
U3	Female	29	6 years	6 (0)	Weekly
U4	Male	24	1 year	2 (0)	Monthly
U5	Male	21	5.5 years	10 (1)	Weekly
U6	Female	20	1 year	2 (0)	Daily
U7	Female	22	1.5 years	2 (0)	Daily
U8	Female	21	5.5 years	21 (0)	Monthly
U9	Male	23	1.5 years	4 (0)	Weekly
U10	Female	39	9 years	9 (3)	Monthly
U11	Female	22	5 years	8 (1)	Monthly
U12	Male	22	2 years	6 (1)	Rarely

Note: Numbers in parentheses under "Written Sci-Fi Works" indicate formally published Sci-Fi works.

7.2 Procedure

We conducted remote user studies on Tencent Meeting and recorded the whole process for analysis. Participants first completed a background survey (5 minutes) to collect information about their science fiction writing experience (Table 3). This was followed by an overview of the study (5 minutes) and a tutorial on using the system (15 minutes), including a walkthrough of the interface, functions, and example operations. Participants accessed the system via Chrome Remote Desktop on the interviewer's computer.

To assess the efficacy of our system in supporting the human–LLM co-ideation experience compared to mainstream writing support tools, we conducted a within-subjects study consisting of two ideation sessions (45 minutes each) with randomized order. In each session, participants were tasked with producing a preliminary outline for a short sci-fi story. Their specific goal was to construct fictional worlds and develop storylines, rather than drafting full narrative text.

One session used Sudowrite's Canvas mode, distinct from its standard editor mode, as the baseline because (i) it is a specialized commercial tool designed for fiction writing; (ii) the Canvas mode features a comparable spatial interface composed of text blocks and visual links; and (iii) the Canvas mode utilizes the same underlying model (GPT-4) as our system. The other session used Sci-Fi Spark. During each ideation session, we encouraged participants to think aloud to learn about their thought processes and reactions.

After each ideation session, participants completed a questionnaire survey (10 minutes) consisting of three sections, each aligned with one research question. Since our system is designed to act as a co-creator, we focused on obtaining self-assessed user experience rather than third-party evaluation of output quality.

Finally, we conducted a semi-structured interview (20 minutes) to gather qualitative insights aligned with the three research questions. Participants were asked: (Q1) How did the generated inspirations from the two generators help you develop ideas? (Q2) How did the Ideation Canvas help you visualize and communicate idea relationships with the LLMs? (Q3) How did the system support your nonlinear, iterative ideation process? To better identify the underlying reasons behind participants' responses, we followed up with probing questions to relate feedback to specific system features. Participants were also encouraged to share any additional feedback. All interviews were audio-recorded and transcribed for analysis. Two researchers independently annotated the transcripts to identify key themes. Disagreements were resolved through iterative discussion, leading to a consensus set of findings.

7.3 Measurement

We combined objective and subjective measures to comprehensively evaluate the system's effectiveness.

7.3.1 Objective Measures. To understand how the system influenced writers' ideation processes, we logged user interaction data and derived three key metrics to quantify the process:

(1) **Generative utilization:** Measured by the number of system-generated idea blocks and the adoption rate.
(2) **Structural connectivity:** Measured by the number of narrative relationships established by users, including total relations, internal consistency relations, and temporal consistency relations.
(3) **Co-creation iteration:** Measured by the frequency of text edits within blocks and modifications to relation lines.

7.3.2 Subjective Measures. Participants completed a questionnaire after experiencing each tool. The survey consisted of three sections aligned with our research questions.

(1) **Creativity Support (Q1):** Assessed participants' perceptions of inspiration (adapted from the Inspiration Scale [76]), novelty (adapted from Novelty metric in Creative Product Analysis Matrix [14]), and consistency, (adapted from Resolution metric in Creative Product Analysis Matrix [14]). All items used a 5-point Likert scale and are shown in Figure 7.
(2) **Structure Visualization Support (Q2):** Assessed participants' perceived mental effort in understanding and controlling idea relationships using the Ideation Canvas. We adapted items from the NASA-TLX scale [40], as it effectively captures cognitive load in visual and interactive tasks. All questions are shown in Figure 8.
(3) **Co-creation Experience (Q3):** Assessed whether the system's usability and the co-creation experience. Items were adapted from the System Usability Scale (SUS) and prior co-creation research [59] that views AI as a co-creator. All questions are shown in the left section of Figure 9.

7.3.3 Data Analysis. For survey data, we conducted Wilcoxon Signed-Rank Tests to compare statistical differences between the baseline and Sci-Fi Spark conditions. For objective measures, paired samples t-tests were employed. For qualitative data gathered from think-aloud protocols and post-study interviews, two researchers applied thematic analysis to identify key themes regarding participants' experiences.

7.4 Results

7.4.1 Creativity support. Sci-Fi Spark was perceived to effectively stimulate novel ideation. As shown in the survey results (Figure 7), participants rated the system significantly higher than the baseline in inspiring writing ideas and providing novel ideas. Qualitatively, participants noted that compared to Sudowrite, which was often described as "generic" or "formulaic" , Sci-Fi Spark offered distinct perspectives. For example, U2 and U6 commented: "[The baseline] didn't really diverge my thoughts... but Sci-Fi Spark offered twists with less 'AI flavor' and truly diverged my thinking."

This ability to stimulate novelty was attributed to generation strategies that mirrored writers' creative thinking patterns. U4 and U12 emphasized that the approach aligned with their natural cognitive processes: "Recombining existing concepts is how we usually think." Among the strategies, the Combination mode was seen as a collaborative mechanism that introduced external stimuli.

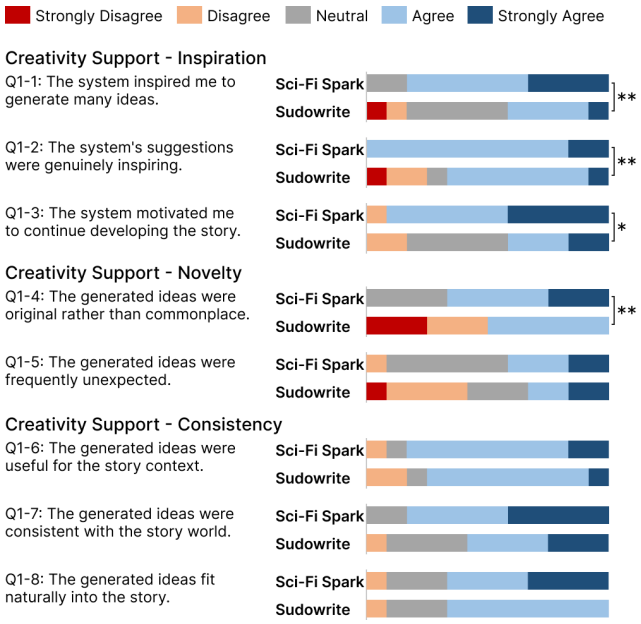


Figure 7: Survey results comparing Sci-Fi Spark and Sudowrite on perceived creativity support, including inspiration, novelty and consistency. ** indicates statistical significance at $p < 0.05$, * indicates marginal significance.

U1 and U5 noted: "It brings in things seemingly unrelated... which breaks fixations." Furthermore, unlike the baseline's automated bulk generation, the system's granular generation enhanced user control. U4 and U7 explained: "I can choose element tags to conceive content according to my own ideas, giving me a higher sense of control."

Regarding consistency, survey results showed no significant difference, indicating that both systems were perceived as capable of generating ideas that fit the context. However, qualitative feedback revealed a critical distinction. Participants noted that while Sudowrite was consistent, it often "relied on fixed templates" (P11) that ignored specific user intents while logical. In contrast, Sci-Fi Spark aligned better with the ongoing story structure and writers' evolving intents. U1, U3, and U11 commented: "It optimized the story points based on my specific requirements... and fits the timeline better."

This alignment was attributed to the step-by-step control during generation. U5 and U11 pointed out that unlike the bulk generation of the baseline, Sci-Fi Spark allowed them to build narrative sequentially: "It generates consistent content following the order... helping me think about the next step." This incremental control allowed the system to move forward with the writer rather than dictating a pre-determined narrative path.

These subjective perceptions are substantiated by objective measures. As shown in Table 4, although the baseline generated more text blocks ($M_{base} = 21.25$, $M_{ours} = 19.25$) because its "Generate Outline" feature produces a dozen blocks in a single click, participants adopted a significantly higher proportion of ideas from Sci-Fi Spark ($M_{base} = 18.10\%$, $M_{ours} = 45.10\%$, $p < 0.05$). This significant contrast confirms that while the baseline produced consistent

Table 4: Comparison of objective measures between Sci-Fi Spark and Sudowrite (N=12). Values are reported as Mean (SD). Statistical significance was determined using paired samples t-tests.

Category	Metric	Sci-Fi Spark	Sudowrite	p-value	Sig.
		Mean (SD)	Mean (SD)		
Generative utility	Generated Blocks	19.25 (5.46)	21.25 (8.20)	.498	-
	Adoption Rate	45.1% (18.1%)	18.1% (13.6%)	<.001	***
Structural connectivity	Total Relations	12.75 (6.89)	4.42 (5.28)	.005	**
	Internal Relations	7.58 (4.48)	0.00 (0.00)	<.001	***
	Temporal Relations	5.17 (2.72)	4.42 (5.28)	.662	-
Co-creation iteration	Relationship Edits	0.83 (1.27)	0.00 (0.00)	.044	*
	Block Edits	3.00 (2.04)	2.42 (4.29)	.664	-

Note: * $p < .05$, ** $p < .01$, *** $p < .001$.

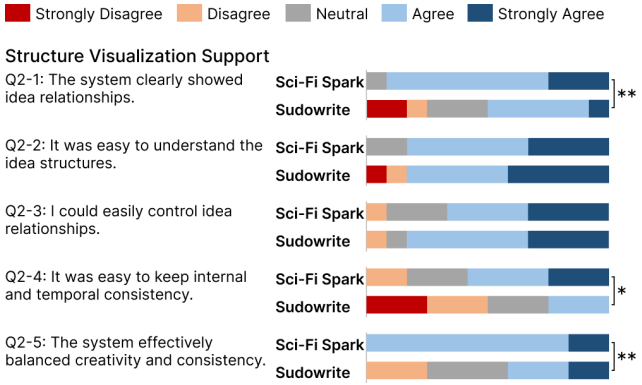


Figure 8: Survey results comparing Sci-Fi Spark and Sudowrite on perceived structure visualization support. ** indicates statistical significance at $p < 0.05$, * indicates marginal significance.

text, its template nature limited its actual utility; conversely, Sci-Fi Spark's high adoption rate proves that its user-aligned consistency was far more valuable to the creative process.

7.4.2 Structure visualization support. Sci-Fi Spark was perceived to provide clearer and more controllable support for managing complex narrative structures. Survey results (Figure 8) indicate significant advantages in clearly showing idea relationships (Q2-1) and balancing creativity with consistency (Q2-5). While both systems were rated similarly for ease of use (Q2-2, Q2-3), Sci-Fi Spark showed a marginal advantage in maintaining internal and temporal consistency (Q2-4).

Qualitative feedback attributes these results to a trade-off between simplicity and adaptibility. Participants found that Sudowrite's linear timeline structure was suitable for simple narratives, but struggled with the complexity inherent to Sci-Fi. P9 noted that the timeline structure became overwhelming when dealing with multiple storylines or world settings: "It became chaotic and hard to understand" . Consequently, participants felt this rigid linearity constrained divergent exploration. In contrast, Sci-Fi Spark's dual-relationship visual structure effectively supported the management of complex idea relationships flexibly. U1 and U5 explained that the flexibility matched their workflow: "It matches the divergent thinking of Sci-Fi... allowing me to backtrack and control complex storylines with richer branches." Furthermore, this structure enabled users to organize fragmented thoughts. U1, U6, and U12 noted: "[I can] unfold the entire story... and organize scattered pieces thoroughly."

These perceptions are substantiated by objective measures. As shown in Table 4, participants created significantly more relationship links in Sci-Fi Spark ($M_{ours} = 12.75$, $M_{base} = 4.42$). Crucially, while the baseline was limited to temporal sequencing, participants actively utilized Sci-Fi Spark's internal consistency links ($M_{ours} = 7.58$, $M_{base} = 0$) to define logical dependencies. This disparity confirms that while the baseline's linear structure is sufficient for basic sequencing, a dual-relationship structure is essential for capturing the multi-dimensional complexity of Sci-Fi co-creation.

7.4.3 Co-creator of nonlinear ideation workflows. Sci-Fi Spark was perceived to offer a more useful and collaborative co-creation experience, particularly in integrating features (Q3-1) enabling writers to express creative goals (Q3-6) (Figure 9). Objective measures further reveal that participants in Sci-Fi Spark group engaged more in structural iteration, with significantly more relationship edits in Sci-Fi Spark ($M_{ours} = 0.83$, $M_{base} = 0.00$, $p < 0.05$). The baseline's zero value reflects its rigid constraints because the timeline structure is predefined and prevents users from modifying connection lines. While text block edits were similar, this contrast highlights a fundamental difference in structural iteration flexibility.

Sci-Fi Spark was seen as well-suited for supporting iterative and nonlinear ideation workflows, as shown in the action logs (Figure 10). Across participants, an ideation session with Sci-Fi Spark typically followed three phases: an early phase dominated by worldbuilding and Novelty invocations, a middle phase of rapid switching where linking and Consistency invocations became frequent as fragments were woven into trajectories, and a late phase focused on local refinement with clusters of linking, Consistency,

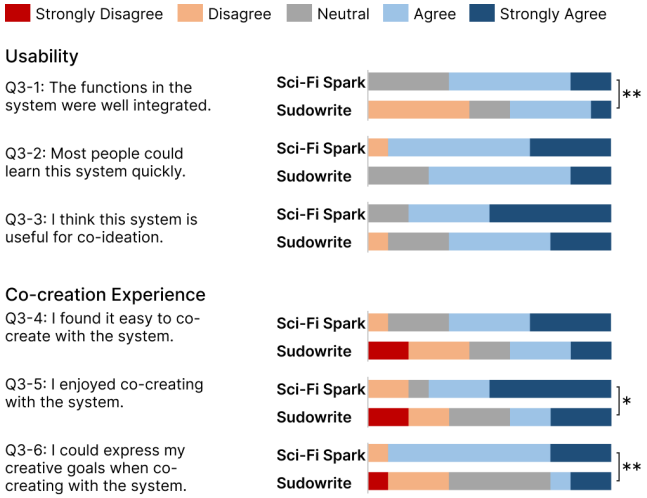


Figure 9: Survey results comparing Sci-Fi Spark and Sudowrite on usability and co-creation experience. ** indicates statistical significance at $p < 0.05$, * indicates marginal significance.

Novelty, and regeneration around specific regions of the canvas. Experts (U2, U5, U10, U11, U12) tended to enter storylining earlier and then briefly return to worldbuilding, and they showed denser, more evenly distributed linking and regeneration events, suggesting continuous maintenance of a story graph; several novices, by contrast, mainly used the Novelty Generator without heavily linking blocks, treating the canvas more as an idea pool than a structure to manage. We also see local cycles where Novelty is used to introduce new blocks and Consistency plus linking are immediately applied to embed them (highlighted by red underlines in Figure 10), indicating that writers orchestrate the two generators in complementary divergence–convergence loops rather than using them in isolation.

8 Discussion

8.1 Canvas-based visual structuring for creative writing

The Ideation Canvas in Sci-Fi Spark serves as an external scaffold of creative cognition. Compared to prior canvas-based ideation systems such as TaleStream [28], our Ideation Canvas not only displays ideas but actively supports nonlinear ideation workflows. It allows writers to record fragmented ideas freely and then progressively organize, connect, and refine them. The structured visualization helps writers detect narrative patterns, identify inconsistencies, and trigger new associations, which is particularly suited to Sci-Fi writers' habits. For instance, one participant noted, "I just realized how these two plots can be connected when they're linked to the same world block." Unlike some systems that require predefined structures, the Ideation Canvas encourages a record-then-organize approach: writers can first capture ideas, and then progressively build relationships flexibly as their ideas evolve. This flexibility aligns with the demands of creative ideation tasks, fostering an interactive loop of externalization and refinement.

Generalizability beyond Sci-Fi. This kind of visual structuring extends beyond Sci-Fi to genres that combine novelty with constraint. In fantasy, for example, writers can use idea blocks to record world settings, character attributes, and key events, then articulate internal consistency links between settings and events and temporal links among events. In historical fiction, blocks can encode period settings, sources, and scene events, while links capture factual constraints and chronological order. In short, the combination of idea blocks with internal and temporal links offers a broadly applicable approach to externalizing and managing narrative structure wherever worldbuilding and storylining co-evolve.

Scalability to larger tasks. Our study focuses on the ideation phase and is presented as a demonstration scenario, where canvases contain a modest number of idea blocks rather than full-manuscript scale. Future work can support larger scale and more complex tasks through grouping and labeling of blocks, layered canvas views, or tag-based filtering. Overall, our findings suggest that hybrid visual–textual representations, tailored to the demands of specific writing tasks and the habits of writers, can better externalize creative cognition. Future creative writing support systems may benefit from integrating visual-textual interfaces, visual structuring, and flexible ideation systems.

8.2 Designing for human–AI collaboration in creative tasks

Our user study found that when a system maintains contextual alignment and interaction fluency, users are more likely to perceive it as a collaborator rather than a tool. This aligns with sociological definitions of collaboration, which emphasize shared context and mutual understanding as foundational to cooperative engagement [45]. The sense of co-creation emerged from contextual alignment enabled by the Ideation Canvas, which reduced the cognitive cost of synchronization. As U5 noted, "The system helps express text relationships to the AI... reducing (my) workload." It also stemmed from seamless interaction that matched the writer's flow: "(I can) freely establish relationships with previous blocks" (U6), allowing the user to maintain their train of thought without interruption. These findings offer empirical insights for HCI research on how interaction design can shape users' perceived sense of collaboration in creative tasks.

In practice, achieving contextual alignment and fluent interaction both require human-centered design. To support contextual alignment, the system should avoid presenting disorganized information that disrupts the user's mental model. As P9 noted regarding the baseline's lack of alignment, "The relationships became chaotic... I had no willingness to iterate." Furthermore, balancing guidance and flexibility is essential for enabling fluent interaction. Systems should avoid overwhelming users with excessive, uncontrolled generation. As U1 warned, "(Generating too many nodes at once) limits inspiration and creates a psychological burden to fill them." Instead, providing controllable mechanisms, such as the ability to "backtrack and control complex storylines" (U5), makes the collaboration feel seamless and supportive.

Beyond maintaining alignment, a significant implication for human-AI collaboration in creative tasks is the necessity of strategic context decoupling. While typical co-creation paradigms prioritize

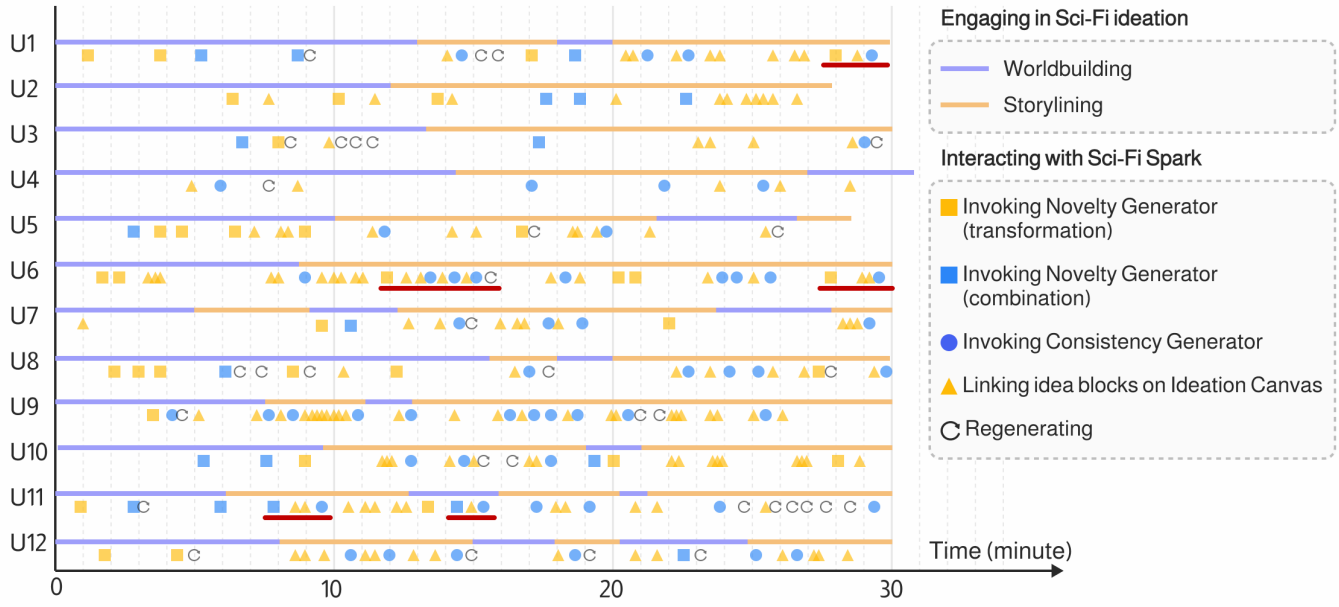


Figure 10: User study action log visualization.

contextual consistency, excessive alignment can lead to creative fixation and predictable model outputs in creativity-intensive tasks. Our study results indicate that writers benefit from temporarily detaching from the existing narrative to generate novel inspirations. This suggests a need for systems that support intentional context detachment, facilitating the injection of external information that is independent of the preceding narrative. For HCI designers, this implies that collaborative agency should include the ability to disrupt the anchoring effect of the established story structure. Such mechanisms help writers maintain the cognitive estrangement essential to the genre by allowing them to pause the contextual flow.

8.3 Limitations and future work

This study has several limitations that inform directions for future work. First, while the Novelty Generator is based on the English-language TVTropes database, participants were Chinese Sci-Fi writers. This mismatch in cultural background may have influenced how participants perceived and interpreted the tropes. To mitigate this issue, we translated and manually curated the trope dataset. Still, future work could integrate multilingual and culturally diverse databases to support broader user populations. Second, our user study focused on short-term ideation sessions due to time constraints. This limits our understanding of long-term usage in two perspectives. From a process perspective, we could not observe how the system supports full-scale writing projects where co-ideation needs evolve over weeks or months. From a cognitive perspective, we were unable to assess how prolonged engagement with such systems affects human writers' creativity development and cognitive patterns. Finally, some participants noted that visual elements, such as concept sketches or mood boards, could complement textual ideas. Integrating multimodal input and generation in future work may further enhance the system's ability to support creativity and assist in describing abstract or speculative concepts.

9 Conclusion

In this paper, we introduce Sci-Fi Spark, a human–AI co-creation system that instantiates an interaction paradigm for supporting the nonlinear and iterative Sci-Fi ideation. By applying the Ideation Canvas as a unified conceptual model, alongside the Novelty and Consistency Generators, the system addresses key challenges in generating novel and consistent inspirations. The technical evaluation and user study show that Sci-Fi Spark enhances creativity, supports narrative structure visualization, and enables fluid collaboration with LLMs. Beyond the specific domain of Sci-Fi, this work offers a generalizable design metaphor for future human-AI interfaces aiming to scaffold complex, multi-dimensional creative workflows.

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